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Multiphysics modeling of the influence of initial pressure on mechanical and electrochemical performance of all-solid-state batteries

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ABSTRACT

All-solid-state batteries (ASSBs) are emerging as promising energy storage solutions owing to their enhanced safety and high energy density. However, the application of higher pressures is also required to improve electrochemical performance. This study employs a multiphysics approach by combining the discrete element method (DEM) and finite element method (FEM) to investigate the impact of the compression rate and initial pressure on electrochemical performance. To the best of the authors' knowledge, this is the first study to integrate the DEM and FEM methodologies for the prediction of electrochemical and mechanical responses, capitalizing on the advantages of both approaches. The DEM is used to investigate the compaction behaviors of solid electrolyte and electrode mixtures, yielding essential data on mechanical properties, which are subsequently incorporated into the FEM model. The FEM is utilized to systematically explore the electrochemical performance and mechanical responses. The findings highlight that elevated stress enhances geometric properties, including surface area and volume fraction, leading to improved electronic and ionic conductivity. Although higher compression ratios can enhance electrochemical performance, they also elevate stress levels, thereby increasing the potential for mechanical failure. Therefore, a well-balanced electrode design is crucial. This study provides valuable insights into the design of ASSBs.

1. Introduction

All-solid-state batteries (ASSBs) have attracted considerable interest as the batteries of the future owing to their enhanced safety features and higher energy density [1]. However, the unique characteristics of solid electrolytes necessitate the application of higher pressures during fabrication to improve the electrolyte–electrode contact, enhance structural integrity, and minimize interfacial resistance [2]. Furthermore, it is important to maintain pressure throughout the charge and discharge reactions during electrode preparation and battery operation. The pressure applied during the fabrication is commonly known as the fabrication pressure, whereas the pressure required during battery operation is referred to as the stack pressure [2].

Several previous studies have focused on the relationship between the fabrication and stack pressures and solid-state battery (SSB) performance [1–7]. For example, Kato et al. [7] successfully fabricated a thick-electrode all-solid-state lithium (Li)-ion battery with 500 MPa of fabrication pressure and 50 MPa of stack pressure. Hansel et al. [3] demonstrated the influence of stack pressure in Li metal SSBs to suppress

the Li metal extrusion. A stack pressure above 5 MPa stopped Li extrusion, and a high fabrication pressure was required to minimize the interfacial contact resistance. Doux et al. [6] analyzed the effects of fabrication and stack pressures on electrolyte conductivity and battery performance. They found that a low operating stack pressure reduced the apparent ionic conductivity owing to poor contact between the electrolyte and current collectors. However, this surprisingly did not detrimentally affect the cyclability of the SSBs. Based on previous literature, the stack pressure was suggested to be less than 1 MPa for practical reasons [8] compared with the commonly used high fabrication pressure of approximately hundreds of MPa [2]. In addition, it has been suggested that achieving optimal contact between the active particles and solid electrolyte interface is crucial. The cell performance can be compromised if the pressure is too low, whereas excessively high pressures can lead to issues, such as cracks or failure of the electrode, that can decrease cell performance.

Modeling of the microstructure of the active particles and solid electrolytes has been previously investigated to examine the contact between the active particles and solid electrolyte interface and

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subsequently improve the performance of ASSBs [9–15]. Froboese et al. [9] developed model electrodes with four active material sizes and six volume fractions to investigate the microstructures and ionic conductivities of ASSBs. Yamakawa et al. [10] investigated the constitutive correlations between the microstructures and power densities of ASSBs. Kimura et al. [12] investigated the influence of active-material loading on electrochemical reactions using computed tomography (CT) combined with X-ray absorption near-edge structure spectroscopy. Three dimensional (3D) microstructures and charging state mapping were acquired by reconstructing the CT images, which provided information regarding the electrochemical reactions. However, the above studies mainly investigated the microstructures of the electrochemical parameters and ASSB performance. A finite element method (FEM)-based mechanical model should also be employed to investigate the effects of fabrication and stack pressure on ASSBs. For example, Lee et al. [15] developed a single-particle model to study the stress generation in active particles surrounded by solid electrolytes at different locations. However, this model only consisted of a single particle and could not fully understand the detailed microstructure consisting of numerous active and solid electrolyte particles. Nunoshita et al. [16] also developed a numerical model by incorporating the volume expansion of anode active materials in all-solid-state Li-ion batteries, which strongly affected the dynamic changes in the electrode structure and its activity in electrochemical reactions. However, this model was also based on the porous electrode theory, which could not incorporate the effect of the microstructures.

It is advantageous to employ bottom-up models, such as molecular dynamics and the discrete element method (DEM), to gain a deeper understanding of how mechanical characteristics affect the microstructure of the electrode [13]. The DEM is particularly advantageous

because it allows for the simulation of individual particles and their interactions, enabling the accurate derivation of a stress–strain profile and direct analysis of stress localization through particle misalignment. This feature is particularly suitable for analyzing ASSBs because they are composed of compressed particles on an individual basis. Accordingly, recent studies [13,14] have employed the DEM to simulate various aspects of ASSB electrode fabrication, including cold-press manufacturing, intercalation damage, aggregate destruction, and particle consolidation. These studies primarily focused on the output parameters related to geometric characteristics, such as porosity, volume fraction, and tortuosity, as well as electrochemical parameters, including ionic conductivity. However, these studies did not fully explore the electrochemical performance of the system.

Therefore, this study utilizes a multiphysics approach combining the DEM and electrochemical–mechanical coupled FEM to examine the impact of initial pressure on the electrochemical performance of ASSBs. To the best of our knowledge, this is the first study to integrate the DEM and FEM methodologies for the simultaneous prediction of electrochemical and mechanical responses in ASSBs, capitalizing on the advantages of both approaches. The DEM component enables the derivation of a realistic compression profile and effective transport properties by considering the behavior of individual particles within the ASSBs. In addition, the FEM component facilitates the investigation of the mechanical and electrochemical performances of the ASSBs. A comprehensive analysis of the behavior of ASSBs under different initial pressures is conducted by integrating these two modeling techniques.

The remainder of this paper is organized as follows. Section 2 introduces the DEM and FEM models used to study mechanical and electrochemical behaviors of ASSBs. Section 3 presents the DEM and FEM results of electrochemical performances and mechanical responses of

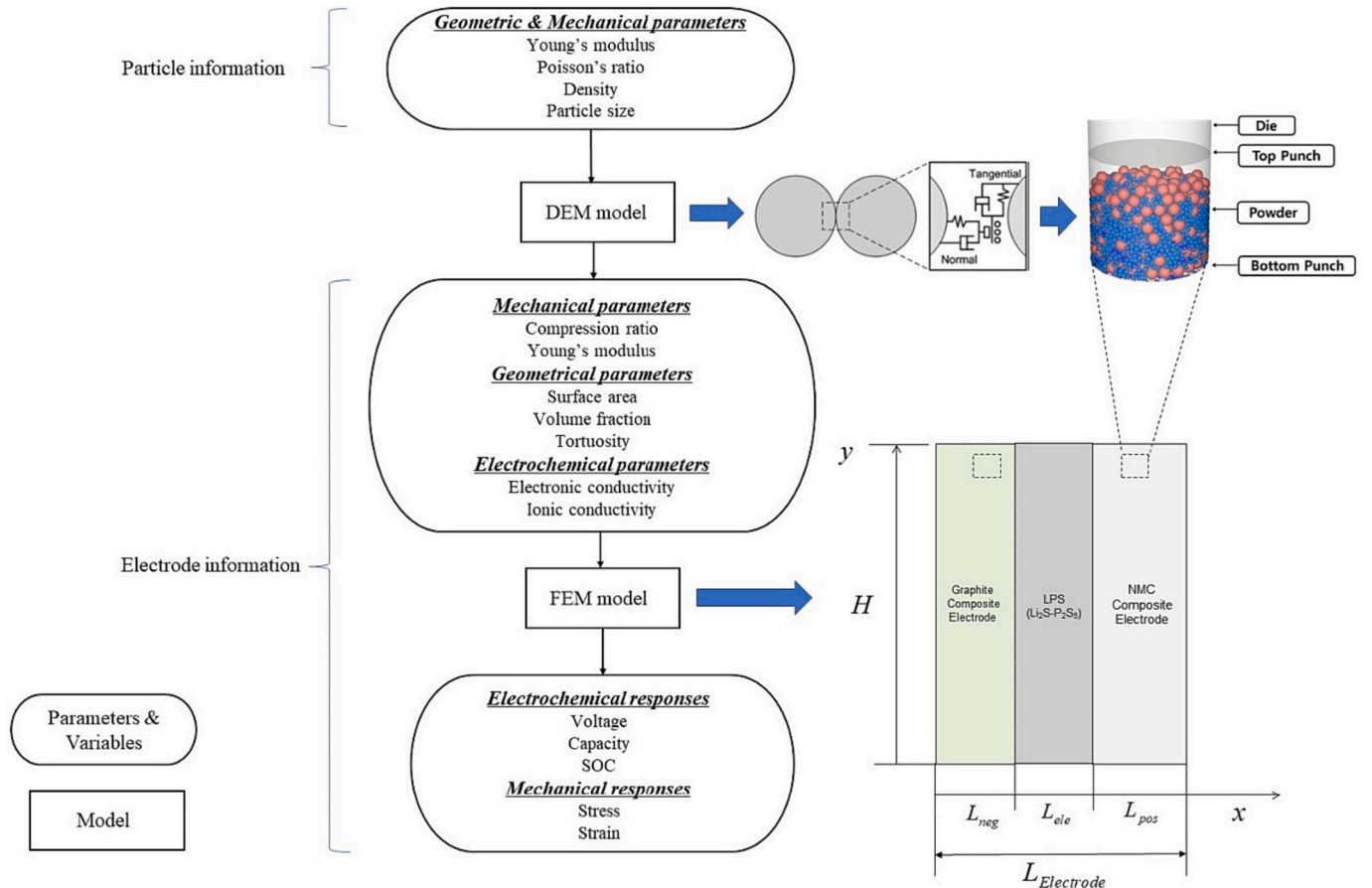


Fig. 1. Overall simulation flow chart and schematics of DEM using contact model and FEM for an ASSB.

ASSBs. Finally, Section 4 summarizes the study and discusses its limitations and future studies.

2. Methods

Fig. 1 illustrates the flow chart for the entire simulation, along with schematics related to the DEM and FEM models. At the particle level, geometric and mechanical parameters serve as inputs to the DEM model. These parameters, including Young's modulus, Poisson's ratio, density, and particle size, can be found in Table 1.

The DEM model is employed to explore mechanisms for compression characteristics at the particulate level. Here, individual particles are considered, and contacts between particles are defined to efficiently analyze powder behavior. The electrochemical and mechanical models are coupled in the following manner. The DEM model outputs mechanical parameters at the electrode level, such as the compression ratio and Young's modulus. The compression ratio influences geometric parameters like volume fraction, facilitating the derivation of surface area from Eq. (34) and tortuosity from Eq. (38). These geometric parameters, in turn, affect electrochemical parameters, influencing electronic and ionic conductivity. Electronic conductivity changes according to Eq. (35), while ionic conductivity is influenced by both mechanical and geometrical factors, as described in Eqs. (36) and (37). Increases in electronic and ionic conductivities will reduce the overall impedance of the cell. These electrochemical changes impact concentrations inside the electrode and the voltages derived from electrochemical equations. Changes in solid-phase lithium concentration, as indicated in Eq. (29), result in volumetric changes. All these alterations simultaneously affect the electrochemical performance and mechanical response characteristics.

The variations in mechanical, electrochemical, and geometrical parameters are input into a coupled Finite Element Method (FEM) model. This integration enables the extraction of electrochemical responses (voltage, concentration, capacity, state of charge, etc.) and mechanical responses (stress, strain) based on the coupled electrochemical-mechanical interactions.

2.1. DEM model

The DEM was utilized to explore the existing mechanisms for compression characteristics at the particulate level, where particles are treated individually and contacts between particles can be defined to efficiently check the behavior of the powders. The powder compaction process of the DEM is shown in the right upper part in Fig. 1 [17].

The powders were loaded into the die through particle generation to ensure a sufficient mixing ratio. Subsequently, the top punch initiated vertical compression, while the bottom punch was fixed. The impulse associated to the contact force in normal and tangential directions are computed as:

$$(J_\alpha)_c = \int_c F_\alpha dt \approx \sum_c F_\alpha \Delta t \quad (1)$$

where $(J_\alpha)_c$ are the normal ($\alpha = n$) and tangential ($\alpha = \tau$) components of the impulse produced by the contact force acting during a complete

collision, F_α is the normal ($\alpha = n$) and tangential ($\alpha = \tau$) component of the contact force and Δt is the time lapse.

In this work, the impulse components are calculated from the contact forces defined by the contact model, i.e. hysteresis linear spring model. The average values of the normal and tangential stresses are computed as:

$$\sigma_\alpha = \frac{\sum_{c=1}^{N_c} (J_\alpha)_c}{A_i \Delta t_{out} N_p^{en}} \quad (2)$$

where σ_α is the normal ($\alpha = n$) or tangential ($\alpha = \tau$) component of the stress, A_i is the area of the region of influence, Δt_{out} is the time lapse between two consecutive outputs, N_c is the total number of complete collisions on the region of influence during output time step Δt_{out} , and N_p^{en} is the number of enabled particles in the given particle group, at the time when the statistics are displayed.

The average stress values are obtained by dividing the total sum of impulses between particles into area of influence and time lapse.

The hysteretic linear spring model was selected as the contact model to ensure sufficient accuracy for compressible materials and directly consider particle contact. The developed models were verified from a previous study [18]. The energy dissipation did not depend on the relative velocities of the particles because viscous attenuation was not considered. Therefore, the energy dissipation was not affected by other contacts. The kinetic equations in the DEM are expressed as:

$$\begin{aligned} F_n^t &= \min(K_{nl} S_n^t, F_n^{t-\Delta t} + K_{nu} \Delta S_n) \quad \text{if } \Delta S_n \geq 0 \text{ or} \\ F_n^t &= \max(F_n^{t-\Delta t} + K_{nu} \Delta S_n, \lambda K_{nl} S_n^t) \quad \text{if } \Delta S_n \leq 0 \end{aligned} \quad (3)$$

$$\Delta S_n = S_n^t - S_n^{t-\Delta t}, \quad (4)$$

where F_n^t and $F_n^{t-\Delta t}$ denote the normal elastic-plastic contact forces at the current t and previous time $t - \Delta t$, respectively. ΔS_n denotes the difference in the contact normal overlap and λ denotes the dimensionless constant. S_n^t and $S_n^{t-\Delta t}$ denote the contact normal overlaps at the current and previous times, respectively. K_{nu} and K_{nl} denote the unloading and loading contact stiffnesses, respectively.

The loading/unloading stiffness is defined by the particle size, elastic modulus, and coefficient of restitution between the two selected contact materials. This coefficient of restitution is a measure of the energy dissipation for a pair of materials. The stiffness of loading for particle-particle and particle-boundary contacts are respectively expressed as:

$$\begin{aligned} \frac{1}{K_{nl}} &= \frac{1}{K_{nl-p_1}} + \frac{1}{K_{nl-p_2}} \\ \frac{1}{K_{nl}} &= \frac{1}{K_{nl-p}} + \frac{1}{K_{nl-b}} \end{aligned} \quad (5)$$

The stiffness of unloading is expressed as:

$$K_{nu} = \frac{K_{nl}}{\epsilon^2} \quad (6)$$

The individual stiffnesses in relation to particles and boundaries are respectively expressed as:

$$K_{nl-p} = E_p L_{par}, \quad (7)$$

$$K_{nl-b} = E_b L_{par}, \quad (8)$$

where L_{par} denotes the particle size, and E_p and E_b denote the Young's moduli of the particles and boundary, respectively.

Powder compaction models were developed for various powder mixtures, including solid electrolytes (Li₇La₃Zr₂O₁₂ (LLZO)/Li₁₀GeP₂S₁₂ (LGPS)), cathode materials (LiNi_xMn_yCo₂O₂ (NMC)/LGPS/carbon black), and anode materials (graphite/LGPS/carbon black). The mechanical properties of the powders used in the DEM analysis are listed in

Table 1
Material properties for powder compaction.

Material	Density (kg/m ³)	Young's modulus (GPa)	Poisson's ratio	Particle size (um)
Graphite [18]	2180	1	0.25	14
NMC [19]	4750	144	0.25	12
LLZO [20]	5500	150.3	0.26	30
LGPS	2000	37.2	0.3	30
Carbon black [21]	1950	80	0.3	4

Table 1.

The coefficient of frictions were set to 0.1 for graphite and carbon black, 0.2 for NMC, and 0.3 for LLZO and LGPS. Spherical particles were utilized, and between 1000 and 30,000 particles were generated to confirm accuracy. The powder filling rate within the die was set to 80%.

The effect of initial positions in mixing configuration for powder compaction has been reported previously that randomly positioned powder mixtures displayed good agreement with the experimental results [17]. Particle generation method in DEM was developed to further enhance reflection of the experimental environment [18].

Determination of the initial position of the particles for the particle generation method involves several steps: first, particles are generated randomly and simultaneously from the identical plain, where slight variations in each generation results could be observed. Then the particles fall down freely, and finally are allowed to settle down under gravity. By considering parameters such as angle of repose, velocity, and contact definition between particles, realistic mixing configurations could be achieved [22,23].

To check the validity of the mixing configurations further, the mixed powders were divided into 8 sections, and the volume ratio of the two different powders for each section and the standard deviation were explored. The average volume ratio/standard deviation was 0.695/0.118 for NMC and 0.682/0.047 for graphite, respectively. Although the variations are noticeable, these values incorporate the behavior of particles, including settling process after positioning.

The software utilized for the DEM analysis was Ansys Rocky DEM 4.5. The hysteresis linear spring model was chosen as the contact model to ensure accuracy for material compression and to directly consider contacts of particles. In addition, the boundary collision statistics module was applied to calculate the stress.

2.2. FEM model

2.2.1. Electrochemical modeling

This study employed a pseudo-3D physics-based model to formulate an electrochemical-mechanical coupled model with two-dimensional boundary conditions for ASSBs. This model was applied to an ASSB encompassing x- and y-axes, as depicted in the right lower part in Fig. 1. A representative active material particle within the positive and negative composite electrodes was investigated.

The solid-phase potential of the electrode was determined by applying a charge-transfer equation governing the behavior of the Li ions within the solid phases. This equation can be expressed as:

$$\frac{\partial}{\partial x} \left(\kappa_s^{\text{eff}} \frac{\partial \phi_s}{\partial x} \right) - a_s i_{Li^+} = 0, \quad (9)$$

where κ_s^{eff} denotes the effective solid-phase conductivity, ϕ_s denotes the solid-phase potential, i_{Li^+} denotes the current density of the Li intercalation and deintercalation reactions, and a_s denotes the surface area of the electrode.

The electronic conductivity of the electrode changes with the volume fraction of the electrode, yielding:

$$\kappa_s^{\text{eff}} = \kappa_s \epsilon_s^{\text{brug}}, \quad (10)$$

where brug denotes the Bruggeman number.

An electrical ground was established at the left end of the negative current collector, and the entire current density was borne by the solid phase at the right end of the positive current collector. This is expressed as:

$$\phi_s = 0 \text{ at } x = 0, \quad (11)$$

$$-\kappa_s^{\text{eff}} \frac{\partial \phi_s}{\partial x} = I_{\text{app}} \text{ at } x = L, \quad (12)$$

where I_{app} denotes the applied current density, which is positive when the battery discharges and negative when the battery charges. The other boundary conditions for the solid-phase potential ϕ_s in the y-direction are expressed as:

$$\frac{\partial \phi_s}{\partial y} = 0 \text{ at } y = 0 \text{ and } y = H \quad (13)$$

Analyses of diffusion-induced variations in Li concentration and the stress arising from the intercalation and deintercalation processes were conducted within this particle. Moreover, the fluctuations in the Li concentration and stress at the particle level were leveraged to explore the mechanical and electrochemical behaviors at the electrode level. Notably, each spatial point within the electrode corresponded to a volume element at the particle level that was comprised of numerous particles incorporating solid-state electrolytes. This work mainly expands upon a 2D physics-based model of a solid-state Li-ion battery developed by Wolff et al. [24] with single-ion conducting electrolytes. Li ions are the only mobile species, and there is no concentration gradient within the electrolyte, which is expressed as:

$$\frac{\partial c_{Li^+}}{\partial t} = 0, \quad (14)$$

where c_{Li^+} denotes the concentration of the electrolyte.

The charge transfer of Li ions through the electrolyte is expressed as:

$$\nabla \cdot \left(\kappa_e^{\text{eff}} \left(\nabla \cdot \phi_e - \frac{2RT}{F} (1 - t_+^0) \left(1 + \frac{d \ln f_{\pm}}{d \ln c_{Li^+}} \right) \nabla \cdot \ln c_{Li^+} \right) \right) + a_s i_{\text{tot}} = 0, \quad (15)$$

where κ_e^{eff} denotes the effective electrolyte-phase conductivity, ϕ_e denotes the electrolyte-phase potential, R denotes the universal gas constant, T denotes the temperature, F denotes the Faraday constant, t_+^0 denotes the Li-ion transference number, $f_{\pm}$ denotes the electrolyte activity coefficient, and i_{tot} denotes the total current density required for the overall reaction. The second term in the above equation becomes null because there is no concentration gradient within the electrolyte. Furthermore, the last term in the equation becomes null within the electrolyte region because no chemical reaction occurs inside the electrolyte.

The total current density can be defined as the combined sum of the current density associated with the Li intercalation and deintercalation reactions, as well as the current density required for the double-layer capacitance. This can be expressed as:

$$i_{\text{tot}} = i_{Li^+} + i_{DL}, \quad (16)$$

where i_{DL} denotes the current density of double-layer capacitance.

The Li intercalation/deintercalation reactions in the negative and positive electrodes are governed by the Butler-Volmer equation, which is expressed as:

$$i_{Li^+} = i_{0,Li^+} \left[\exp \left(\frac{\alpha_{a,Li} F}{RT} \eta_{Li^+} \right) - \exp \left(-\frac{\alpha_{c,Li} F}{RT} \eta_{Li^+} \right) \right], \quad (17)$$

where $\alpha_{a,Li}$ and $\alpha_{c,Li}$ denote the anodic and cathodic transfer coefficients of the electrochemical reaction, respectively. η_{Li^+} represents the overpotential for Li intercalation and deintercalation reactions and is expressed as:

$$\eta_{Li^+} = \phi_s - \phi_e - U_{Li^+}, \quad (18)$$

where U_{Li^+} denotes the equilibrium potential of the Li intercalation/deintercalation reactions.

For the composite electrode, i_{0,Li^+} represents the exchange current of the Li intercalation/deintercalation reactions, which is expressed as:

$$i_{0,Li^+} = F k_{Li^+}^{0.5} c_{s,\text{surf}}^{0.5} (c_{s,\text{max}} - c_{s,\text{surf}})^{0.5} c_{Li^+}^{0.5}, \quad (19)$$

where k_{Li^+} denotes the reaction rate constant of Li, $c_{s,\text{max}}$ denotes the

maximum concentration of Li ions on the positive electrode, and $c_{s,surf}$ denotes the concentration of Li ions on the surface of the positive electrode particles.

The current density depends on the derivative of the interfacial voltage, which can be expressed as [25]:

$$i_{DL} = a_s C_{DL} \frac{\partial(\phi_s - \phi_e)}{\partial t}, \quad (20)$$

where C_{DL} denotes the double layer capacitance.

The boundary conditions for the electrolyte potential ϕ_e are expressed as:

$$\frac{\partial \phi_e}{\partial x} = 0 \text{ at } x = 0 \text{ and } x = L_{neg} + L_{ele} + L_{pos}, \quad (21)$$

$$\frac{\partial \phi_e}{\partial y} = 0 \text{ at } y = 0 \text{ and } y = H \quad (22)$$

The material transport of the Li ions inside the active particle is expressed as:

$$\frac{\partial c_s}{\partial t} + \nabla \cdot \mathbf{J} = 0, \quad (23)$$

where c_s denotes the concentration of Li ions in the solid phase, and $\mathbf{J}$ denotes the flux of the Li-ion species.

The boundary conditions for the material transport equation can be expressed as:

$$-D_s \frac{\partial c_s}{\partial r} \Big|_{r=0} = 0, \quad (24)$$

$$-D_s \frac{\partial c_s}{\partial r} \Big|_{r=r_p} = J = \frac{i_{Li^+}}{F}, \quad (25)$$

where r_p denotes the radius of the active particle.

A constant flux of Li ions was directly correlated with the current density and applied to the particle surfaces. This Li-ion flux diffused radially from the surface toward the center of the particle, yielding a flux value of zero when the radial distance reached zero.

2.2.2. Mechanical modeling

It is imperative to integrate mechanical deformation to account for intertwined electrochemical-mechanical phenomena. This can be accomplished by using:

$$\nabla \cdot \boldsymbol{\sigma} + \mathbf{b} = 0, \quad (26)$$

where $\boldsymbol{\sigma}$ denotes the macroscopic stress tensor and $\mathbf{b}$ denotes the vector of the body forces. The macroscopic stress σ_{ij} at the electrode level is expressed as:

$$\sigma_{ij} = C_{ijkl} (e_{kl} - e^0 \delta_{kl}), \quad (27)$$

where C_{ijkl} denotes the effective elastic modulus, e^0 denotes the volumetric strain, and δ_{kl} denotes the Kronecker delta. The effective elastic modulus refers to a measure of the stiffness or elasticity of a material when it is subjected to deformation or stress. Young's modulus is a fundamental mechanical property that characterizes how a material deforms under axial or tensile loading. The effective Young's modulus takes into account the influence of various factors, such as the microstructure or composite nature of a material, and provides a representative value that reflects its overall elastic behavior in a specific context or under certain conditions. Consequently, this study utilizes the effective elastic modulus as Young's modulus changes with applied pressure. The effective elastic modulus is expressed as:

$$C_{ijkl} = \frac{E\nu}{(1+\nu)(1-2\nu)} \delta_{ij} \delta_{kl} + \frac{E}{2(1+\nu)} (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk}), \quad (28)$$

which mainly consists of the Young's modulus and Poisson's ratio of the electrode and solid electrolyte, respectively. Detailed derivation of the effective elastic modulus can be found in the supplementary information.

To accommodate the volumetric alterations in the active particles due to Li-ion intercalation/deintercalation, the volumetric strain (ϵ) was defined as the product of the partial molar volume of the active particle and changes in Li concentration, which is expressed as:

$$e^0 = e^0_{xx} = e^0_{yy} = e^0_{zz} = \frac{\Omega \epsilon_s}{3} (c_{s,avg} - c_{s,init}) \quad (29)$$

The boundary conditions of Eq. (26) are expressed as:

$$u_x = u_y = 0 \text{ at } x = 0, \quad (30)$$

$$\sigma_x = P_{ext} \text{ and } \sigma_y = 0 \text{ at } x = L, \quad (31)$$

$$u_y = 0 \text{ at } y = 0 \text{ and } y = H, \quad (32)$$

where P_{ext} denotes the external pressure exerted on the outside from the electrodes. A fixed boundary condition was applied to the left end.

2.2.3. Model parameters

The solid electrolytes used in this study were LGPS and LLZO, and the active anode and cathode materials were graphite and NMC. These active materials were combined with solid electrolytes to create composite bulk electrodes for ASSBs. The key parameters employed in this model are listed in Table 2.

Important mechanical properties, especially the Young's modulus, and the volume fractions of the anode, cathode, and solid electrolyte that varied with compression were determined using the DEM. The required stress increased continuously as the compression progressed, leading to a gradual increase in the Young's modulus of the entire electrode. Additionally, the volume fractions of the anode, cathode, and solid electrolyte increased with the progression of compression, whereas the volume fraction of the voids steadily decreased.

The radius of the active material underwent corresponding changes during Li intercalation and deintercalation as a consequence of the volume fluctuations in the active material. Consequently, alterations in the volume fraction and radius of the active material led to modifications in the surface area, which are expressed as:

$$R_p = (1 + e(c_{AM}/c_{AM,max}))^{1/3} R_{p,0}, \quad (33)$$

$$S_a = 3 \frac{\epsilon_{AM}}{R_p}, \quad (34)$$

where $R_{p,0}$ denotes the initial radius of the active material.

The electronic conductivities of oxide- and sulfide-based solid electrolytes are not negligible. For example, the electronic conductivities of LLZO solid electrolytes at 30 °C were reported as $5.5 \times 10^{-8} \text{ S cm}^{-1}$, which was much lower than those of Li phosphorus oxynitride solid electrolytes and separators used in conventional Li-ion batteries [43]. The electronic conductivity of a solid electrolyte can be influenced by external pressure, especially in the case of low-density solid electrolyte pellets. The proposed model incorporated Maxwell's equation to account for this effect, which is widely used for calculating the conductivity of sintered powder compacts with porosity and is expressed as [44]:

$$\kappa_{AM} = \kappa_{AM,0} \frac{2(1-\theta)}{2+\theta}, \quad (35)$$

where $\kappa_{AM,0}$ denotes the electronic conductivity of a fully dense solid electrolyte pellet without any voids, and θ denotes the compact porosity of a solid electrolyte pellet, which ranges from 0 to 1.

In a previous investigation, the ionic conductivity of a solid electrolyte was determined to be dependent on the operational stack pres-

Table 2

Major physical, electrochemical, and mechanical parameters used in the model.

Parameter	Anode (graphite)	Solid electrolyte (LGPS)	Cathode (NMC)
Geometry			
Layer thickness L	65 μm	75 μm	70 μm
Electrode height	975 μm	975 μm	975 μm
$H_{\text{electrode}}$			
Particle radius R_p	10 μm [26]		5 μm [26]
Initial volume fraction of active material $\varepsilon_{AM,0}$	0.32 [27]		0.18 [2]
Initial volume fraction of solid electrolyte $\varepsilon_{SE,0}$	0.24 [27]	1	0.38 [2]
Initial volume fraction of void $\varepsilon_{v,0}$	0.44 [28,29]		0.44 [2]
Kinetics			
Reaction rate constant of Li k_{Li^+}	$5.03 \times 10^{-12} \text{ m}^{2.5} \text{ s}^{-1} \text{ mol}^{-0.5}$		$1.31 \times 10^{-11} \text{ m}^{2.5} \text{ s}^{-1} \text{ mol}^{-0.5}$
Charge transfer coefficient α	0.5		0.5
Double layer capacitance C_{DL}	0.19 F m^{-2} [30]		0.49 F m^{-2} [30]
Transport			
Initial Li concentration of active material $c_{s,0}$	0.9 $c_{s,max}$ (discharge)		0.1 $c_{s,max}$ (discharge)
Maximum Li concentration of active material $c_{s,max}$	0.1 $c_{s,max}$ (charge) 26,797 mol m^{-3} [26]		0.9 $c_{s,max}$ (charge) 49,886 mol m^{-3} [26]
Diffusion coefficient of active material D_s	$5.0 \times 10^{-12} \text{ m}^2 \text{ s}^{-1}$		$5.0 \times 10^{-12} \text{ m}^2 \text{ s}^{-1}$
Diffusion coefficient of solid electrolyte D_{SE}		$2.61 \times 10^{-4} \text{ m}^2 \text{ s}^{-1}$ [31]	
Li transference number t_{Li}^0	0.363 [32]	0.363 [32]	0.363 [32]
Electrolyte activity coefficient $f_{\pm}$	1 [33]	1 [33]	1 [33]
Electrical Conductivity κ_1	100 S m^{-1} [34]	$2.2 \times 10^{-7} \text{ S m}^{-1}$ [35]	10 S m^{-1} [36]
Electrolyte properties			
Li concentration of solid electrolyte c_{SE}	38,400 mol m^{-3} [37]	38,400 mol m^{-3} [37]	38,400 mol m^{-3} [37]
Mechanics			
Young's Modulus of electrode E_{ele}	Varies depending on compression rate		Varies depending on compression rate
Young's Modulus of solid electrolyte E_{SE}	Varies depending on compression rate	Varies depending on compression rate	Varies depending on compression rate
Poisson's ratio of active particle ν_{AM}	0.32 [38]		0.25 [39]
Poisson's ratio of solid electrolyte ν_{SE}		0.29 [40]	
Partial Molar Volume Ω	$5.19 \times 10^{-6} \text{ m}^3 \text{ mol}^{-1}$ [41,42]		$3.5 \times 10^{-6} \text{ m}^3 \text{ mol}^{-1}$

sure [45]. The following semi-empirical equation was derived to accommodate this effect by extrapolating from the aforementioned experimental data [45]:

$$\sigma_{SE} = \sigma_{SE,max} - (\sigma_{SE,max} - \sigma_{SE,0}) \exp(-k_3 E_{ext}), \quad (36)$$

where $\sigma_{SE,0}$ and $\sigma_{SE,max}$ denote the initial and maximum ionic conductivities, respectively. The empirical constant k_3 was extracted from the previous experimental data [45]. σ_0 , σ_{max} , and k_3 were determined to be 0.002 S cm^{-1} , 0.0002 S cm^{-1} , and 70 GPa^{-1} , respectively. This equation characterizes the asymptotic behavior of LGPS ionic conductivity with respect to stack pressure, as reported in previous study [45].

In contrast to the ionic conductivity of a solid electrolyte pellet on its own, the effective ionic conductivity within solid composite electrodes varies according to the volume ratio and tortuosity of the solid electrolyte within the electrode [46,47]. The estimation of the effective ionic conductivity within the composite electrode considers the ionic conductivity and volume ratio of the solid electrolyte along with the path resistance coefficient of the entire electrode, which is expressed as [46]:

$$\sigma_{SE,eff} = \frac{\sigma_{SE} \varepsilon_{SE}}{\tau_{L/S-all}}, \quad (37)$$

$$\tau_{L/S-all} = -11.7 \varepsilon_{SE} + 8.54 \quad (38)$$

In this context, the empirical equation that models the path resistance coefficient α is derived from a prior investigation, which assumes a linear correlation between tortuosity and the volume fraction of the solid electrolyte, as detailed in previous study [46]. Simulations were conducted for eight cases to compare the electrochemical and mechanical response characteristics based on the external pressure and compression ratios of the cathode, anode, and solid electrolyte, as listed in Table 3.

Achieving the desired electrode configuration with a specific fabrication pressure for the anode, electrolyte, and cathode in a single manufacturing step poses a challenge in solid-state battery design. To address this limitation, a common practice in solid-state battery fabrication involves individually compressing each electrolyte, anode, and cathode component to the desired thickness before assembling them. Alternatively, some fabrication processes first apply pressure to two components before conducting an overall compression. This approach is widespread in the field, serving as a solution to the difficulty of achieving the desired electrode design with a single, uniform fabrication pressure. Support for this practice is evident in prior research. [48–50] For instance, $\text{Li}_{6.4}\text{Ga}_{0.2}\text{La}_3\text{Zr}_2\text{O}_{12}$ ceramic electrolyte powder was initially processed under 300 MPa at room temperature for 10 min [49]. In another study [48], the cathode powder was pressed onto the $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$ electrolyte pellet under 555 MPa. The anode, utilizing Li foil, did not undergo a separate compression process. Eventually, the three layers of the cathode/electrolyte/anode were pressed together at 19 MPa to fabricate the all-solid-state batteries. A pressure of 230 MPa was applied to the battery during the charge-discharge operation. By individually compressing each anode and cathode component, it becomes possible to apply the desired fabrication pressure or compression rate to each part. The assembly of these compressed components then subjects the entire system to stack pressure. Since the fabrication pressure and stack pressure (external pressure) are utilized differently,

Table 3

Simulation case studies with different external pressures and different compression rates of the negative electrode, positive electrode, and solid electrolyte.

Case#	External pressures (MPa)	Compression rates of negative electrodes (anode)	Compression rates of solid electrolytes	Compression rates of positive electrodes (cathode)
1	10	0.7	0.7	0.7
2	10	0.5	0.5	0.5
3	10	0.7	0.5	0.5
4	10	0.5	0.5	0.7
5	10	0.5	0.7	0.5
6	0	0.7	0.7	0.7
7	50	0.7	0.7	0.7
8	100	0.7	0.7	0.7

allowing for individual control, the methodology employed in this study, where each component undergoes compression according to the desired compression ratio or fabrication pressure and applies different stack pressure, closely mirrors real-world situations. This approach enables the investigation of both the electrochemical performance and mechanical response characteristics of the individual components under real-world conditions.

The compression ratio is defined as the ratio of the volume change to the volume before compression. Cases 1 and 2 were compared under the same external pressure conditions, with compression ratios of 0.7 and 0.5, respectively. Cases 3 to 5 were divided to observe the electrochemical and mechanical response characteristics under the same external pressure conditions, with only one compression ratio set to 0.7 in each case. Finally, Cases 6 to 8 involved simulations with fixed compression ratios while varying the external pressure to study the influence on the electrochemical and mechanical characteristics.

The capacity of the electrode is typically designed to be larger for the negative electrode than for the positive electrode. As a result, the overall capacity is determined by the capacity of the positive electrode. The electrode's capacity is calculated by multiplying the volume fraction, representing the volume of the actual electrode in the total volume of both electrodes, by the amount of lithium the electrode can hold at its maximum, multiplied by Faraday's constant, as below

$$Q = L_{pos} \cdot H \cdot W \cdot F \cdot C_{s,max,pos} \cdot E_{s,pos} \quad (39)$$

where Q is the total capacity of the electrode and W is the depth of the electrode. Relative capacity refers to the proportion of capacity currently held at the electrode compared to the total theoretically available capacity Q .

The SOC of each electrode can be defined as the proportion of lithium currently held compared to the maximum amount of lithium the electrode can accommodate, as below

$$SOC_{pos} = \frac{C_{s,pos}}{C_{s,max,pos}}, SOC_{neg} = \frac{C_{s,neg}}{C_{s,max,neg}} \quad (40)$$

The FEM simulation was conducted using COMSOL Multiphysics 6.0 software. Specifically, the mechanical modeling employed the solid mechanics module within the software and was coupled with the detailed electrochemical model described in Section 2.2.1.

3. Results and discussion

3.1. DEM results

3.1.1. Solid electrolyte mixture

Two materials with different mechanical properties (LLZO and LGPS) were used to explore the powder compaction characteristics of a solid electrolyte mixture, which can be found in Fig. 2. TP and BP refer to the stress applied to the Top Punch and Bottom Punch, respectively. In practice, stress is applied at the top punch to achieve compression, and this stress is then transmitted through the bulk materials to the bottom stress below. The compressibility of LGPS was significantly higher than that of LLZO when a constant stress was applied to the punch, as shown in Fig. 2(a). Information regarding the variations in Young's modulus and volume fraction was obtained based on the compression rate and stress, and then applied to the FEM model. The Young's modulus exhibited a pattern similar to that of the stress variation.

The compaction responses for the various mixing ratios are shown in Fig. 2(b). It was confirmed that although a greater insertion ratio of high-compressibility material (LGPS 33 at approximately 66 wt%) was important for improving the compaction characteristics, the addition of the high-compressibility material itself significantly enhanced the compaction characteristics. This highlighted the importance of utilizing powder mixtures for solid electrolytes, particularly for materials with high compressibility.

3.1.2. Cathode and anode mixtures

The compaction response of the cathode mixture was also investigated. NMC was chosen as the active material for the positive electrode, and LGPS was used as the solid electrolyte. The composition of the selected cathode materials were NMC (60 wt%), LGPS (35 wt%), and carbon black (5 wt%). The top-punch stress reached the yield stress of the powders at a compression rate of 29%, as shown in Fig. 3(a). At this point, the stress transfer from the top to bottom punch was 51%. In addition, comparisons were made with only the NMC model, where the cathode mixture exhibited better compaction characteristics, as shown in Fig. 3(b). This was due to the particle size difference in the mixture that introduced larger particle-created spaces, even though the yield stresses were similar (approximately 100 MPa). This led to the increased behavior of particles, resulting in improved compaction characteristics.

Graphite was chosen as the active material of the negative electrode and LGPS was chosen as the solid electrolyte for the anode mixture. The compositions of the selected anode materials were graphite (60 wt%), LGPS (38 wt%), and carbon black (2 wt%). A significant improvement in

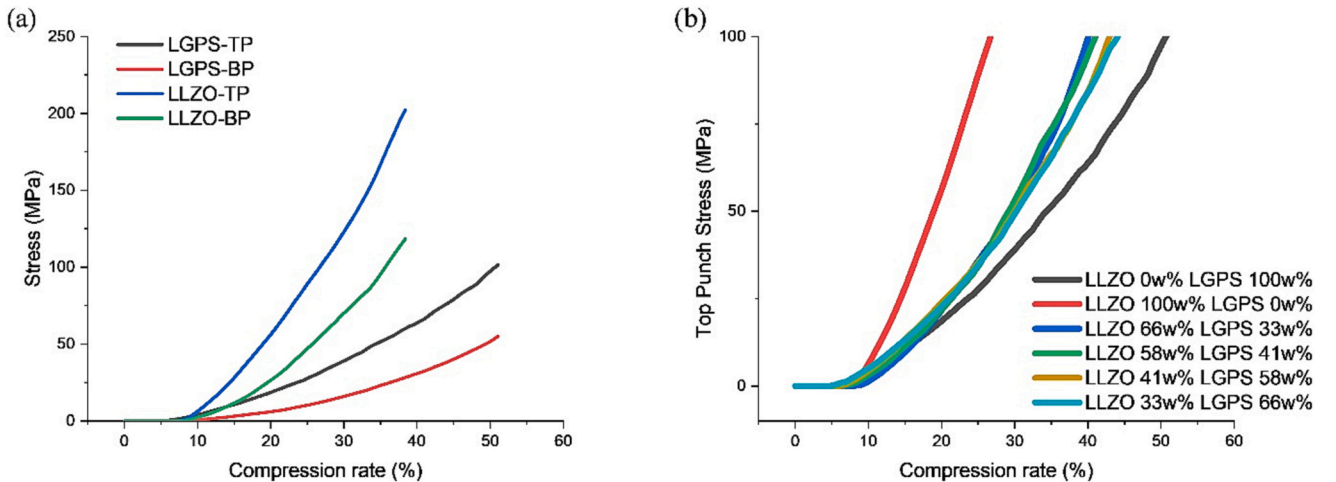


Fig. 2. Graphs of powder compaction for solid electrolyte mixtures: (a) compressibility when a constant stress was applied to the Top punch (TP) and bottom punch (BP). (b) Compaction responses for the various mixing ratios.

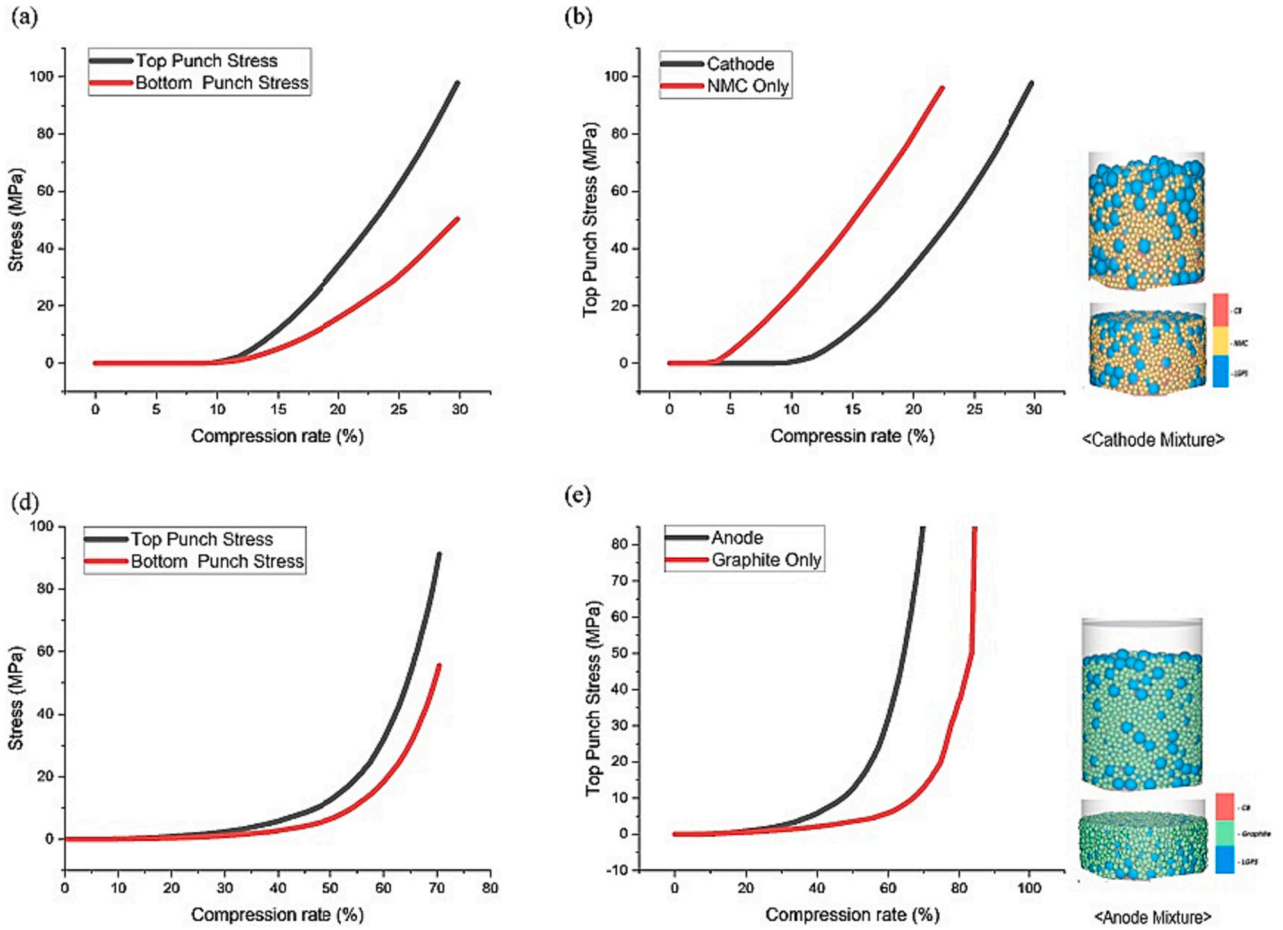


Fig. 3. Plots of punch stress for: (a) cathode mixture, (b) cathode mixture and NMC only, (c) anode mixture, and (d) anode mixture and graphite only.

the compaction response was observed, and a compression rate of 70% and stress transfer of 61% were obtained at the yield stress of the mixed material, as shown in Fig. 3(c). This was mainly because graphite has the characteristics of a solid lubricant with a very high compressibility. Further enhancement was achieved by increasing the graphite content, as shown in Fig. 3(d). However, this led to imperfect compactness. Excessive insertion can lead to a high dispersion of stress owing to the poor ductility of graphite, resulting in incomplete compaction.

3.2. FEM results

3.2.1. Effect of the compression ratio of the anode, cathode, and solid electrolyte

An electric-chemo-mechanically coupled FEM model was employed utilizing precise input values obtained from the microstructure variations due to compression using DEM, such as the electrode volume fraction based on the compression ratio and Young's modulus derived from stress under different compression rates. This approach enabled the prediction of the electrochemical performance and mechanical responses. The discharge voltage variations in an ASSB when compressing the cathode, anode, and solid electrolyte with 70% and 50% compression ratios are compared in Fig. 4(a). A compression ratio of 0.7 exhibited significantly better electrochemical performance. The voids within the electrode decreased as the compression ratio of the electrode increased. In addition, the volume fractions of the cathode, anode, and solid electrolyte increased, leading to significant differences in capacity and voltage. This observation was also evident in the variations of the state of charge (SOC) at four regions within the ASSB: the left part of the anode, middle part where the anode meets the right part of the solid

electrolyte, middle part where the cathode meets the left part of the solid electrolyte, and far-right portion of the cathode, as shown in Fig. 4(b). The same colors and symbols represent the same locations and cases, respectively. The SOC decreased at the cathode and increased at the anode as discharging occurred. The smaller volume fraction of active materials caused a higher rate of SOC change when the same current flowed through the electrode. The volume fraction decreased accordingly when the compression ratio was low, leading to a significant variation in the SOC. Consequently, the slope of the SOC changed sharply, and the difference in the SOC between the cathode and anode increased significantly. As a result, the battery reached the 3.5 V cut-off voltage more quickly, resulting in a lower relative capacity.

The charge voltage difference in an ASSB when compressing the cathode, anode, and solid electrolyte with 70% and 50% compression ratios is shown in Fig. 4(c). A larger capacity was required during charging than during discharging because the coulombic efficiency was lower than 100%, which was caused by factors such as low conductivity and double-layer capacitance. Similar to the discharge, higher compression ratios resulted in increased charging capacity, ultimately demonstrating better electrochemical performance. The SOC decreased in the anode and increased in the cathode during charging, as shown in Fig. 4(d). A higher compression ratio led to a higher volume fraction, resulting in smaller SOC variations and ultimately confirming an increased capacity, similar to Fig. 4(b).

The various von Mises stresses and displacements are depicted in Fig. 5. The von Mises stress and displacement during discharge for Case 1 from Table 3 where the compression ratios of the cathode, anode, and solid electrolyte were set to 0.7 are depicted in Fig. 5(a) and (b), respectively. In this study, von Mises stress is employed to consider the

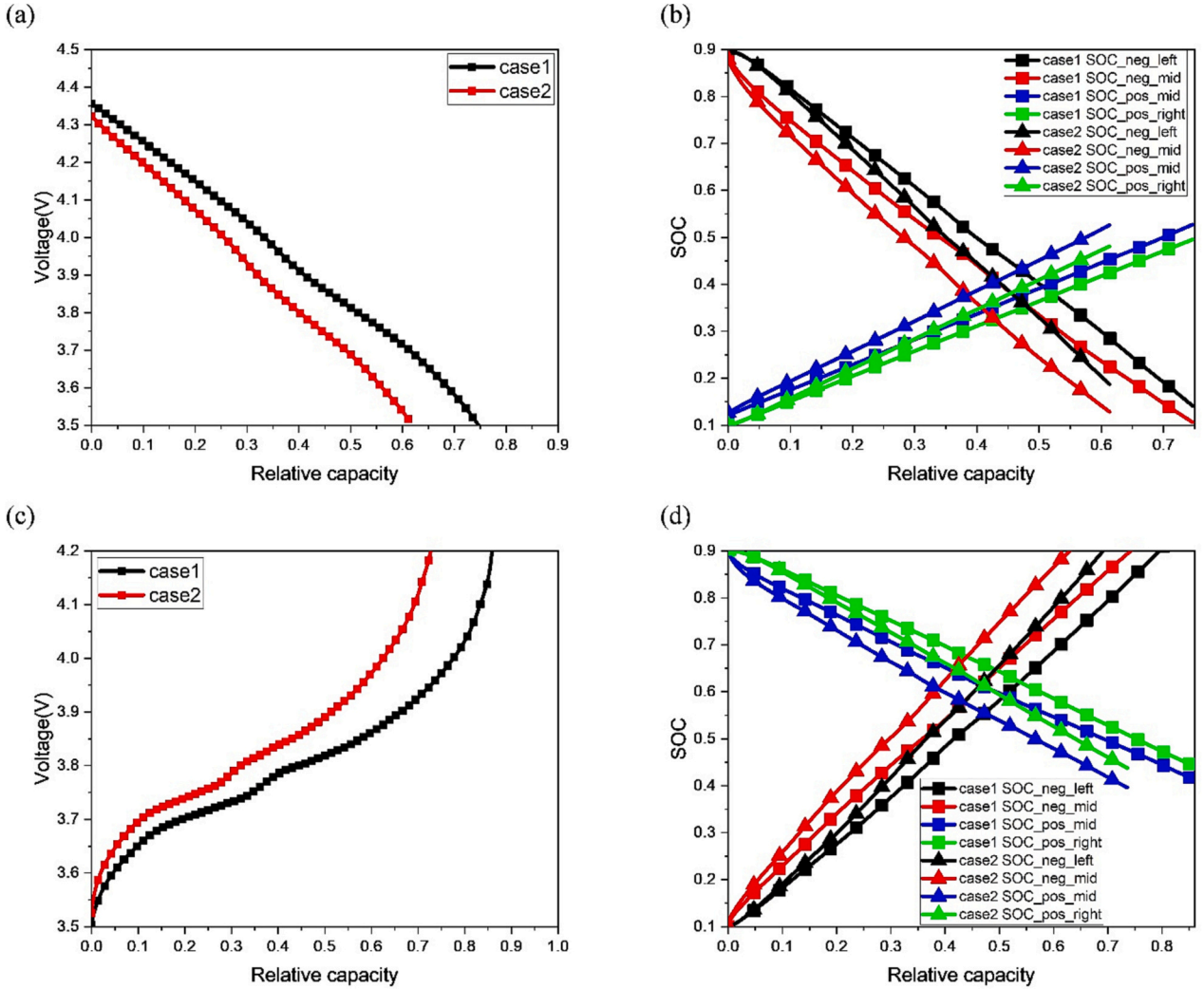


Fig. 4. Plots showing the relative capacity with different compression ratios of electrodes and electrolytes for: (a) voltage during discharge, (b) SOC during discharge, (c) voltage during charge, and (d) SOC changes during charge. In Case 1, the compression rates for the negative electrodes, solid electrolytes, and positive electrodes (cathode) are all set at 0.7, accompanied by an external pressure of 10 MPa. Meanwhile, in Case 2, the compression rates for these components remain at 0.5, and the external pressure is also 10 MPa.

overall deformation characteristics of the material. Previous studies in the field of solid-state batteries also have extensively employed von Mises stress for numerous analyses [51–54]. Specifically, when analyzing the onset and progression of the plastic behavior of electrodes in solid-state batteries, the von Mises yield criterion has been utilized [53]. Additionally, to investigate how stress may contribute to mechanical degradation, an examination of von Mises stress has been conducted [51].

The stress was highest at the cathode, followed by the anode and solid electrolyte. Previous DEM results showed that the stress and Young's modulus were highest in the cathode, followed by the solid electrolyte and anode. However, in terms of volume expansion, the graphite material of the anode exhibited the highest expansion, followed by that of the cathode. The solid electrolyte exhibited no volume expansion. The von Mises stress distribution in a fully discharged state at a compression ratio of 0.7 is shown in Fig. 5(c). The cathode exhibited the most significant expansion in the fully discharged state, whereas the anode contracted. Nevertheless, the stress levels were most pronounced at the cathode, followed by the anode and separator section.

The von Mises stress, displacement, and stress distribution during the discharging and charging processes are shown in Fig. 5(a), (b), and (c) and (d), (e), and (f), respectively. The magnitudes and orders of stress

and displacement in the cathode, anode, and solid electrolyte were similar but with slightly larger absolute values. This was attributed to the relative capacity being greater during charging than that during discharging, leading to a larger Li concentration available for intercalation in the active material. Consequently, the stress and displacement increased accordingly. The most significant expansion and contraction were observed at the cathode and anode for the fully discharged and charged states, as shown in Fig. 5(c) and (f), respectively.

The von Mises stress and displacement during discharge in Case 2 from Table 3 are depicted in Fig. 5(g) and (h), respectively. Here, the compression ratios of the cathode, anode, and solid electrolyte were set to 0.5. The overall stress was much lower than that in Case 1, where the corresponding compression ratios were set to 0.7. The reason for the reduction in the overall von Mises stress was that the compression ratio was smaller at 0.5 compared to that at 0.7, leading to a decrease in the Young's modulus. Moreover, the overall stress was highest in the cathode and lowest in the solid electrolyte, as shown in Fig. 5(a). This was because the Young's modulus of the cathode remained the largest, whereas the expansion ratios of the cathode and anode remained the same. The displacement pattern shown in Fig. 5(h) exhibited a trend very similar to that shown in Fig. 5(b), albeit with slightly smaller absolute values. The lower compression ratio of 0.5 made the system more

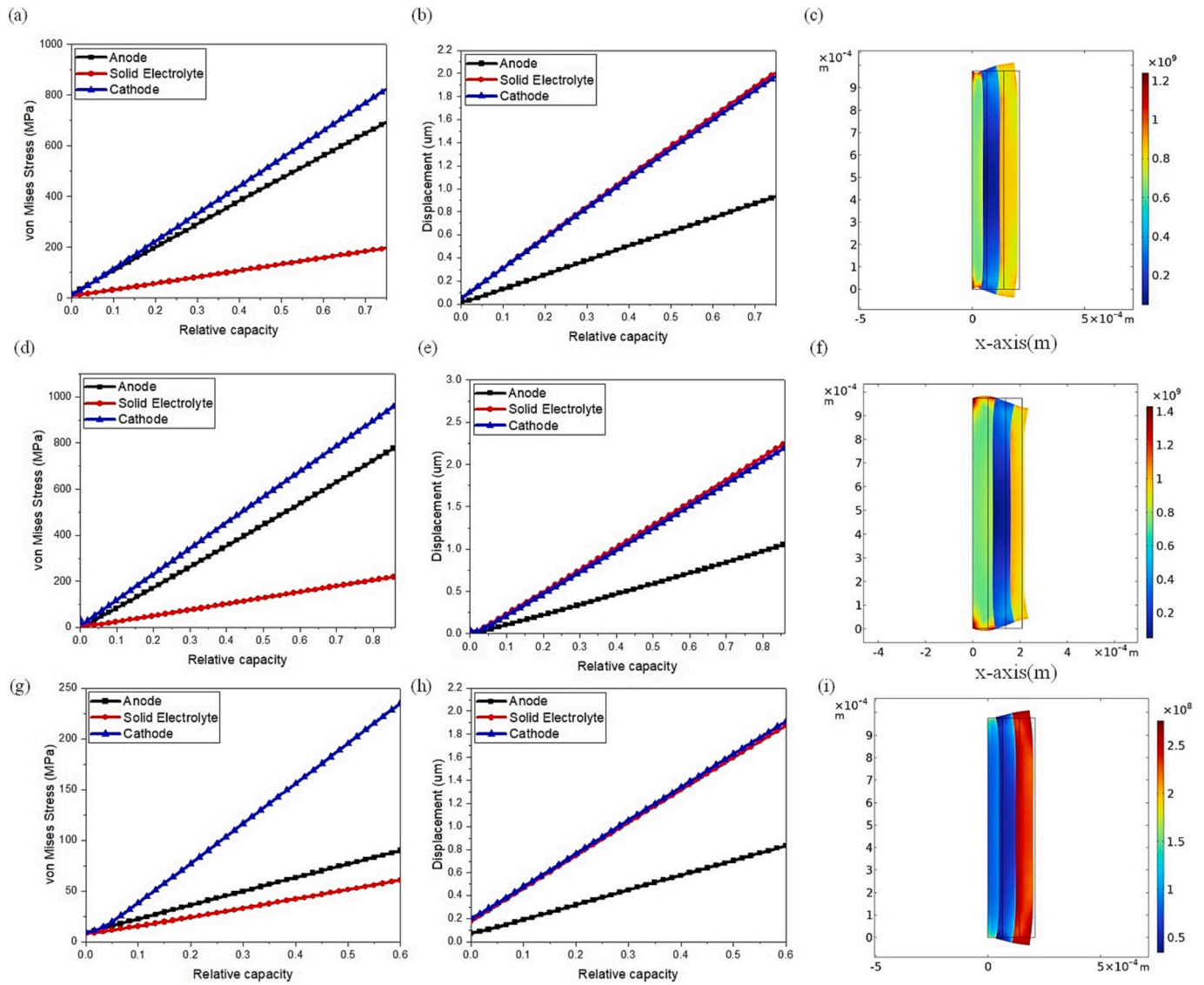


Fig. 5. The von Mises stresses and displacement profiles in the regions of the negative electrode, solid electrolyte, and positive electrode with a compression ratio of 0.7 during: (a) and (b) discharge, and (d) and (e) charge. The von Mises stress distributions with a compression ratio of 0.7 at: (c) fully discharged and (f) fully charged states. (g) von Mises stress and (h) displacement profile in the regions of the negative electrode, solid electrolyte, and positive electrode with a compression ratio of 0.5 during discharge. (i) The von Mises stress distribution at a fully discharged state with a compression ratio of 0.5.

sensitive to external pressure despite applying the same expansion ratios to the cathode, anode, and external pressure, which resulted in a slightly smaller overall displacement. The von Mises stress distribution at a fully discharged state with a compression ratio of 0.5 is shown in Fig. 5(i). The cathode exhibited the most significant expansion and highest distributed stress in the fully discharged state, whereas the anode contracted.

A significant difference in the relative capacity was observed between the compression ratios of 0.7 and 0.5, as shown in Figs. 4 and 5. Compression ratios of 0.7 and 0.5 yielded relative capacities of 0.8 and 0.6, respectively. However, the von Mises stress exhibited a contrasting trend. The maximum stresses occurring during discharge in the anode for compression ratios of 0.7 and 0.5 reached approximately 688 and 76 MPa, respectively. Subsequently, increasing the compression ratio led to enhanced electrochemical performance. However, it also resulted in significantly higher pressure and stress within the electrodes, which can lead to the occurrence of cracks or failure inside an electrode structure due to high pressure. This pressure escalation is caused by reversible reactions due to the intercalation and deintercalation of Li, and irreversible phenomena such as Li plating that occur over long cycling

periods. Therefore, it is essential to consider electrochemical performance and mechanical response characteristics when designing electrodes. These results underscore the importance of accounting for both aspects of electrode design to ensure safe and efficient performance in battery applications.

The compression ratios of each region of the battery were set to 0.7 or 0.5 to investigate the effect of varying the compression ratios in different regions on the overall electrochemical performance and mechanical response characteristics, as shown in Table 3. Specifically, the compression ratios of the anode, cathode, and solid electrolyte in Cases 3, 4, and 5 were set to 0.7, respectively.

Increasing the compression ratio of the cathode achieved the most positive impact on the electrochemical characteristics, whereas the solid electrolyte was the least affected by changes in the compression ratio, as shown in Fig. 6(a). The volume fraction of the active material increased with an increasing compression ratio, and the cathode showed the most significant improvement in electrochemical performance because it was the most sensitive to this change. However, the solid electrolyte area was unaffected by changes in compression ratio and thus showed little

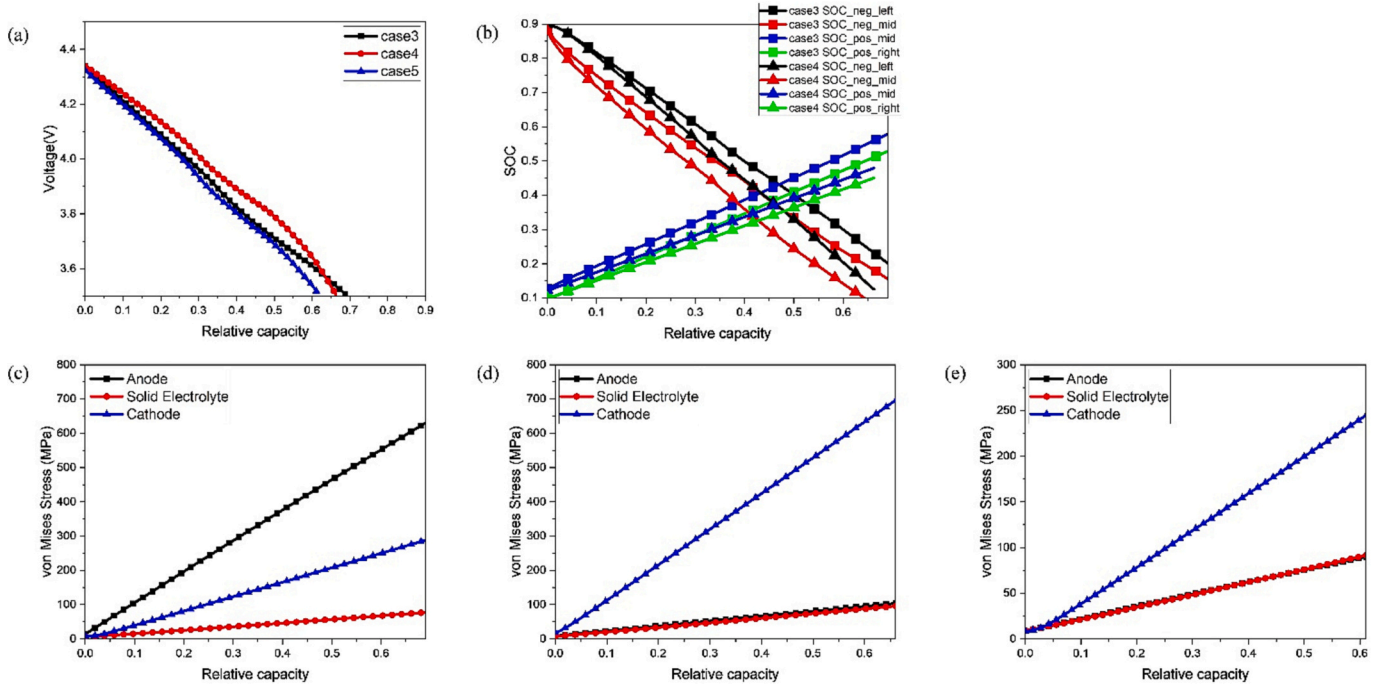


Fig. 6. (a) Voltage profile and (b) SOC changes with respect to the relative capacity with different compression ratios of electrodes and electrolytes during discharge for Cases 3, 4, and 5 in Table 3. The von Mises stress in the regions of the negative electrode, solid electrolyte, and positive electrode during discharge in Cases: (c) 3, (d) 4, and (e) 5. In Case 3, the compression rate for the negative electrodes is 0.7, while the compression rates for both the solid electrolytes and positive electrodes are set at 0.5. Similarly, in Case 4, the compression rate for the positive electrodes is 0.7, while the compression rates for the solid electrolytes and positive electrodes are both set at 0.5. In Case 5, the compression rates for the negative electrodes and positive electrodes are 0.5, while the compression rate for solid electrolytes is set at 0.7. In all Cases for 3, 4, and 5, an external pressure of 10 MPa is applied.

impact on the electrochemical properties. To understand this further, the SOC variations for Cases 3 (increased anode compression ratio) and 4 (increased cathode compression ratio) were plotted, as shown in Fig. 6 (b). Case 3, with a higher anode compression ratio, showed a shallower slope in the graph, leading to earlier termination owing to its smaller overall capacity compared to that of Case 4. Conversely, the graph for the cathode in Case 4 is positioned below and shows a steeper slope, resulting in an earlier termination of the SOC variation. Overall, these results reveal the significance of the compression ratio, especially in the cathode area, in influencing the electrochemical performance. Furthermore, the results demonstrated how changes in the compression ratio could impact the SOC variations differently in all regions of the battery.

The compression ratio in each electrode region affected the electrochemical characteristics and naturally influenced the mechanical response characteristics. The variations in the von Mises stresses during discharge in Cases 3, 4, and 5 for each electrode and electrolyte region are shown in Fig. 6(c), (d), and (e), respectively. Cases 3 (Fig. 6(c)) and 4 (Fig. 6(d)) exhibited the highest stresses at the cathode and anode, respectively. This indicated that the corresponding Young's modulus values increased in cases with higher compression ratios, leading to higher stress levels in those regions. Consequently, the stresses at the anode in Cases 3 and cathode in Case 4 reached approximately 629 and 697 MPa, respectively. Notably, Case 4 exhibited the best electrochemical performance even though it demonstrated the least favorable mechanical response. Finally, in the solid electrolyte region, the stress increased slightly as the compression ratio increased compared to that of the anode. However, similar to the electrochemical properties, it exhibited the least variation under the influence of the compression ratio.

In conclusion, the compression ratio in each electrode region demonstrated a significant effect on the electrochemical and mechanical response characteristics. While higher compression ratios may improve the electrochemical performance, they also lead to higher stress levels

that may increase the risk of mechanical failure in certain regions, highlighting the need for a balanced approach in electrode design. Additionally, the solid electrolyte area appeared to be less affected by compression ratio changes in terms of electrochemical and mechanical characteristics.

3.2.2. Effect of external pressure

The compression ratios of the anode, cathode, and solid electrolyte were fixed at 0.7 while the external pressure was set to 0, 10, 50, and 100 MPa to investigate the influence of external pressure. Cases 6, 7, and 8 represent variations in the external pressure at 0, 50, and 100 MPa, respectively, while maintaining the same compression ratios as in Case 1, as shown in Table 3. The electrochemical performance improved as the external pressure increased, as shown in Fig. 7(a). However, the absence of external pressure led to a significant decrease in cell performance, with marginal improvement beyond 50 MPa. Increasing the external pressure from 0 MPa to only 10 MPa significantly enhanced cell performance. Additional analyses were conducted for Cases 6 and 7 at 0 and 50 MPa, respectively, to investigate these changes further. The variations of SOC in the anode and cathode regions with respect to the changes in the external pressure are shown in Fig. 7(b). In the case with zero external pressure, the slope changes were minimal in the anode's far-left and cathode's far-right regions. However, significant SOC variations occurred at the interfaces between the anode, cathode, and solid electrolyte. This was due to the extremely low external pressure that resulted in a significantly lower ion conductivity in the solid electrolyte. Consequently, there was a large difference in the Li concentration between the anode and cathode. In addition, the Li ions did not adequately penetrate the solid electrolyte interface, which led to a localized voltage increase and rapid termination of the discharge. In contrast, the difference in SOC between the anode and cathode was minimal at an external pressure of 50 MPa, leading to a much higher relative capacity. The SOC difference between the electrodes manifested differently compared with

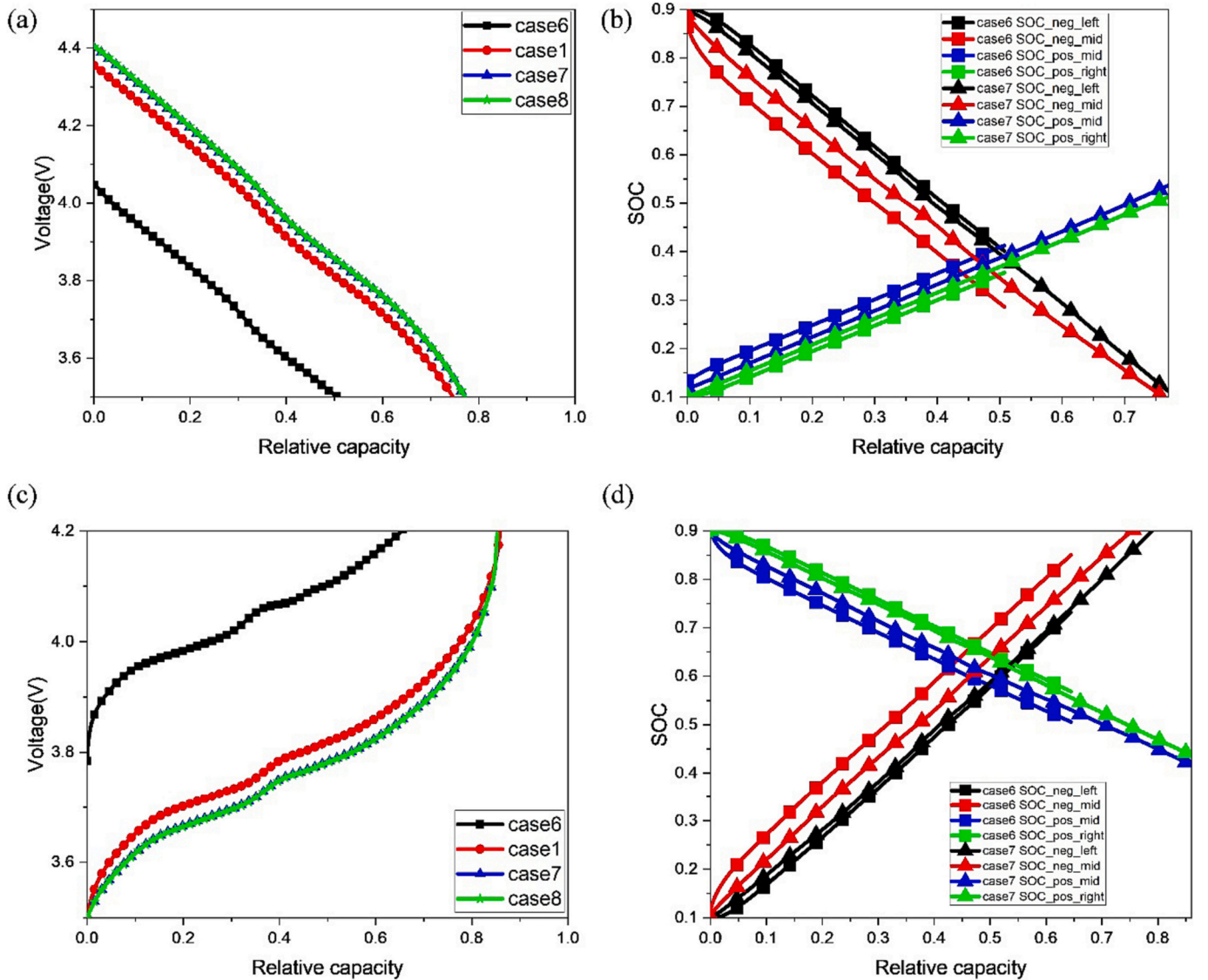


Fig. 7. Voltage profile with respect to the relative capacity with different external pressures during: (a) discharge and (c) charge. SOC changes with respect to the relative capacity with different external pressures during: (b) discharge and (d) charge.

the slope variations of the SOC with respect to the compression ratio shown in Fig. 4(b). The voltage and SOC variations during the charging process are shown in Fig. 7(c) and (d), respectively. These figures demonstrate opposite behaviors to those shown in Fig. 7(a) and (b) during discharge.

The von Mises stress, displacement, and stress distribution in the absence of external pressure and for an external pressure of 50 MPa are shown in Fig. 8(a), (b), and (c), and (d), (e), and (f), respectively.

The external pressure increased when a pressure of 50 MPa was applied, in contrast to the absence of external pressure. However, each region also experienced a significantly higher pressure. For example, the anode experienced a pressure level of 468 MPa in the absence of external pressure, whereas it increased to approximately 739 MPa with an external pressure of 50 MPa. This increase in pressure was attributed to the higher external pressure and increased Li concentration available for intercalation. This subsequently resulted in higher expansion rates at the anode and cathode, which led to higher pressure generation. Furthermore, the overall displacement increased significantly with the applied external pressure, as shown in Fig. 8(b) and (e). The overall deformation pattern in Fig. 8(f) was much more pronounced than that in Fig. 8(c), indicating that a much larger deformation occurred owing to the higher

pressure.

In summary, this study highlighted that the external pressure significantly affected the electrochemical performance, and even a small increase in the external pressure from zero could substantially improve cell performance. The localized SOC variations within the electrodes at a low external pressure demonstrated the importance of carefully considering the external pressure and compression ratio in the electrode design to achieve optimal battery performance.

Parameters related to mechanical behavior were accurately obtained through the developed DEM simulation, including the volume fraction changes and Young's moduli in the cathode, anode, and solid electrolyte based on the applied compression. Moreover, the electrochemical and mechanical behaviors were investigated in combination using the FEM by utilizing the parameters obtained from the DEM. This approach facilitated a better understanding of the trade-off relationship between the electrochemical performance and mechanical response, enabling new solutions that maximized the electrochemical characteristics of the electrodes while remaining within the yield stress limit. The design of electrodes and batteries was optimized by conducting comprehensive research using these methods that harmonized electrochemical performance with mechanical response. This contribution can lead to the

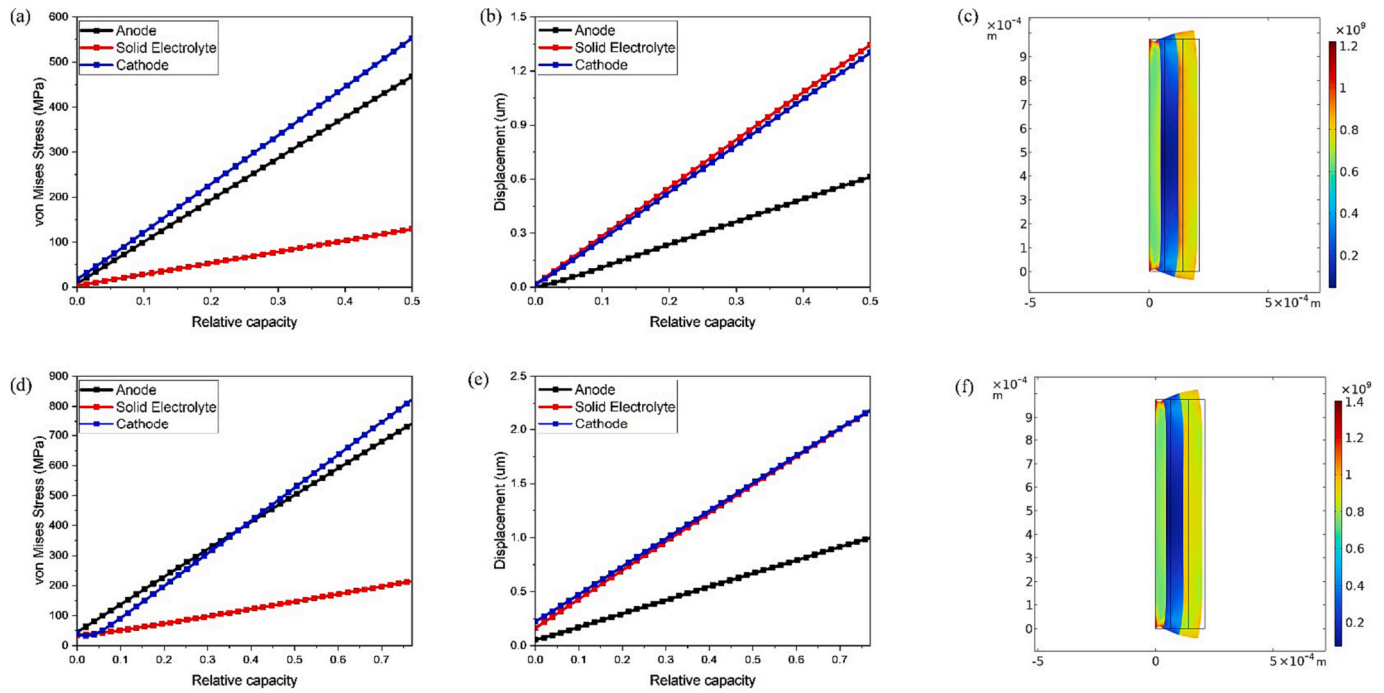


Fig. 8. The von Mises stresses during discharge with: (a) zero external pressure and (d) 50 MPa external pressure. Displacement profiles in the regions of the negative electrode, solid electrolyte, and positive electrode with: (b) zero external pressure and (e) 50 MPa external pressure. The von Mises stress distributions at a fully discharged state with: (c) zero external pressure and (f) 50 MPa external pressure.

development of safer and more efficient batteries, significantly affecting the advancement of the battery industry and innovations in energy-storage technology.

4. Conclusions

This study adopted a multiphysics approach integrating the DEM and electrochemical-mechanical FEM to comprehensively investigate the impact of the initial pressure on the electrochemical performance of ASSBs. Realistic compression profiles and transport properties were modeled employing the DEM that considered the behavior of individual particles within the ASSBs. Simultaneously, the FEM facilitated the exploration of mechanical and electrochemical aspects, enabling a holistic analysis under varying compression rates and initial pressures. To the best of our knowledge, this is the first study to integrate the DEM and FEM methodologies for the simultaneous prediction of electrochemical and mechanical responses in ASSBs, capitalizing on the advantages of both approaches.

The DEM investigations focused on the solid-electrolyte compaction of materials with different mechanical properties. Notably, LGPS exhibited significantly higher compressibility than that of LLZO, resulting in enhanced compaction. Furthermore, the compaction responses of the cathode and anode mixtures were examined, yielding essential data on the Young's modulus and volume fraction for subsequent FEM modeling.

High stress generally enhances overall electrochemical performance. Elevated pressure increases the volume fraction of the electrode's active material, leading to a rise in surface area. This, in turn, facilitates more vigorous electrochemical reactions. Additionally, high pressure increases the volume fraction of the solid electrolyte, improving tortuosity and, consequently, enhancing ionic conductivity. Furthermore, the increase in the volume fraction of the overall solid phase reduces porosity, increasing electronic conductivity.

To assess the impact of external pressure, negative electrode, solid electrolyte, and positive electrode compression rates on electrochemical performance and mechanical response characteristics, an analysis was

conducted by varying the compression ratios using FEM, as shown in Table 3. When the compression ratio of both electrodes and the solid electrolyte was changed from 0.5 to 0.7, the relative capacity increased from 0.6 to 0.8. However, along with this improvement, the maximum stress in the anode increased from 76 MPa to 688 MPa. When comparing the effects of changing the compression ratio for each component, the positive electrode proved to be the most sensitive. Even with the same compression ratio changes, the positive electrode exhibited the best electrochemical performance, despite demonstrating the least favorable mechanical response. External pressure not only increases overall electrode stress but also enhances electrochemical performance, increasing the internally acceptable lithium concentration and consequently raising internal pressure. For example, in the absence of external pressure, the anode experienced a pressure level of 468 MPa, but with an external pressure of 50 MPa, it increased to approximately 739 MPa.

While the mechanisms associated with high external pressure can improve electrochemical performance, it is crucial to consider that increased pressure within the electrode can lead to electrode damage. Therefore, an optimized all-solid-state battery design, considering both electrochemical performance and mechanical response characteristics, is necessary. This study can serve as guidance in designing the electrode for solid-state batteries.

The limitations of this study do not account for abrupt deformations or fractures occurring in the electrode at pressures beyond the yield stress in the Finite Element Method (FEM). To address phenomena such as fracture within the electrode at pressures exceeding the yield stress, these will be considered in future electrode models. By incorporating these factors, we intend to predict issues occurring at the electrode level and develop an optimization framework that can predict the maximum electrochemical performance at stresses below the yield stress in the electrode. This will enable us to provide recommendations for the optimal fabrication pressure.

The optimization of electrode and battery designs can be achieved through this comprehensive research that effectively harmonized the electrochemical performance with the mechanical response. Such advancements have the potential to foster the development of safer and

more efficient batteries, drive innovation in energy-storage technology, and benefit the broader battery industry.

CRediT authorship contribution statement

Yoon Koo Lee: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. **Chaeun Sung:** Data curation, Investigation, Methodology, Visualization. **Jiyeon Kim:** Data curation, Investigation, Methodology, Visualization. **Chaemin Hong:** Data curation, Formal analysis, Investigation, Visualization. **Jinnil Choi:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.est.2024.110431>.

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- A_i : Area of the region of influence, m²
 $brug$: Bruggeman constant
 c_{Li^+} : Concentration of electrolyte, mol m⁻³
 c_s : Concentration of Li ions in the solid phase, mol m⁻³
 $c_{s,max}$: Maximum concentration of Li ion in positive electrode, mol m⁻³
 $c_{s,surf}$: Concentration of Li ions on the surface of the positive electrode particles, mol m⁻³
 $c_{AM,0}$: Concentration of initial Li ion of active material, mol m⁻³
 C_{DL} : Capacitance of double layer, F
 C_{ijkl} : Effective elastic modulus, Pa
 C_{SE} : Concentration of Li ions of solid electrolyte, mol m⁻³
 D_{AM} : Diffusion coefficient of active material, S m⁻¹
 D_{SE} : Diffusion coefficient of solid electrolyte, S m⁻¹
 E_{AM} : Young's modulus of active particle, Pa
 E_{SE} : Young's modulus of solid electrolyte, Pa
 E_p : Young's modulus of particles, Pa
 E_b : Young's modulus of boundary, Pa
 e^0 : Volumetric strain
 F : Faraday's constant, C mol⁻¹
 F_α : normal ($\alpha=n$) or tangential ($\alpha=\tau$) component of the contact force, N
 F_n : Normal elastic-plastic contact forces at current time t , N
 $F_n^{\Delta t}$: Normal elastic-plastic contact forces at previous time $t - \Delta t$, N
 $f_\pm$: Electrolyte activity coefficient
 H : Layer height, m
 I_{app} : Applied current density, A m⁻²
 i_{0, Li^+} : Exchange current of Li intercalation and deintercalation reactions, A m⁻²
 i_{DL} : Current density of double-layer capacitance, A m⁻²
 i_{Li^+} : Current density of Li intercalation and deintercalation, A m⁻²
 i_{tot} : Total current density required for the overall reaction
 J : Flux of Li ions, mol m⁻²s⁻¹
 $(J_c)_i$: Impulse produced by the contact force acting during a complete collision, N s
 k_3 : Empirical constant
 κ_1 : Electrical conductivity, S m⁻¹
 $\kappa_{AM,0}$: Electronic conductivity, S m⁻¹
 k_{Li^+} : Reaction rate constant of Li in the electrodes, m² s⁻¹
 $\kappa_{eff}^{Li^+}$: Effective electrolyte phase conductivity, S m⁻¹
 κ_{eff}^s : Effective solid phase conductivity, S m⁻¹
 K_{unl} : Unloading contact stiffness, N
 K_{nl} : Loading contact stiffness, N
 L : Length of layer, m
 L_{par} : Particle size, m
 N_c : Total number of complete collisions on the region of influence during output time
 N_p^{en} : Number of enabled particles in the given particle group
 P_{ext} : External pressure, Pa
 Q : Total capacity of the electrode, A s
 R : Universal gas constant, J mol⁻¹ K⁻¹
 $R_{p,0}$: Initial radius of the active material, m
 S_n^+ : Contact normal overlap at current time t , m
 $S_n^{t-\Delta t}$: Contact normal overlap at previous time $t - \Delta t$, m
 r_p : Active particle radius, m
 T : Temperature, K
 t_0^+ : Li ion transference number
 U_{Li^+} : Equilibrium potential of Li intercalation/deintercalation reactions, V
 W : Depth of the electrode, m
 α : Charge transfer coefficient
 $\alpha_{a, Li}$: Anodic transfer coefficient of Li intercalation and deintercalation reaction
 $\alpha_{c, Li}$: Cathodic transfer coefficient of Li intercalation and deintercalation reaction
 δ_{kl} : Kronecker delta
 η_{Li^+} : Overpotential for Li intercalation and deintercalation reactions, V
 λ : Dimensionless constant
 σ_α : Normal ($\alpha=n$) or tangential ($\alpha=\tau$) component of the stress, Pa
 σ_{ij} : Macroscopic stress, Pa
 $\sigma_{SE,0}$: Initial ionic conductivity, S m⁻¹
 $\sigma_{SE,max}$: Maximum ionic conductivity, S m⁻¹
 ν_{AM} : Poisson's ratio of active particle
 ν_{SE} : Poisson's ratio of solid electrolyte
 ϕ_e : Potential of electrolyte phase, V
 ϕ_s : Potential of solid phase, V
 $\tau_{L/S,all}$: Resistance coefficient
 θ : Compact porosity
 Ω : Partial molar volume of Li ion in the active material, m³ mol⁻¹
 Δt_{out} : Time lapse between two consecutive outputs, s

Glossary

a_s : Surface area of the electrode, m⁻¹